1. Purpose

The purpose of this study is to evaluate the cytotoxicity of a test article extract using an *in vitro* mammalian cell culture test.

2. Testing Guidelines

2.1. Based on the requirements of the International Organization for Standardization 10993-5, Biological evaluation of medical devices – Part 5: Tests for *in vitro* cytotoxicity.

2.2. Based on Guidance for Industry and FDA Staff, "Use of International Standard ISO 10993-1, "Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process", VI. Test-Specific Considerations, Section

- Temperature: 37C
- Duration: 24 – 72 hours
- Vehicle: allow for extraction of both polar and nonpolar constituents from the test article
    - E.g. mammalian cell culture media (e.g., MEM) supplemented with 5-10% serum

3. GLP Compliance

Good Laboratory Practice – This nonclinical laboratory study will be conducted in accordance with the United States Food and Drug Administration Good Laboratory Practice Regulations, 21 CFR Part 58.

4. Identification of Test and Control Articles

4.1. Test Article

The sponsor, <company name> will submit the test articles, <trade name>, to be evaluated.

4.2. Extraction Vehicle

Single strength Minimum Essential medium supplemented with 5% fetal bovine serum, 2% antibiotics and 1% L-glutamine (1X MEM)

The extraction conditions shall attempt to exaggerate the clinical use conditions so as to define the potential toxicological hazard; however, they should not in any instance cause physical changes such as fusion or melting, which results in a decrease in the available surface area. A slight adherence of the pieces can be tolerated.

Note: the 1X MEM extraction method is conducted in the presence of serum to optimize extraction of both polar and non-polar components.

4.3. Control Articles

5. Test system

5.1. Test System and Justification

5.2. Test System Management

6. Method

6.1. Test Article Preparation

6.2. Control Article Preparation

6.3. Extract Observation

6.4. Test Procedure

7. Evaluation and Statistical Analysis

8. Report

The final report will include information on the cell line, culture medium methods, test and control results, and any additional pertinent observations.

9. Quality Assurance

Inspection will be conducted at intervals adequate to assure the integrity of the study in conformance with 21 CFR 58.35(b)(3). The final report will also be reviewed for conformance to Section 58.185, subpart J, of the GLP Regulations.

10. References